

Electric Revolution 2026: Resilience — India: Big Ambition, Borrowed Supply Chain

Mapping India's EV Supply-Chain Self-Sufficiency Across Batteries, Motors, Minerals, Electronics and Policy

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TL;DR

- **India's EV demand is real and accelerating but its supply chain is not: EVs hit 2,452,921 units (up 25% YoY) and c.8.3% penetration in FY26, yet ~100% of lithium/cobalt/nickel, c.75% of cells and c.93% of NdFeB magnets are still imported, mostly from China.** India today is a battery *pack-assembler* and vehicle-integrator sitting on top of a Chinese upstream. Traction battery demand is heading from c.20 GWh in FY26 to c.95–100 GWh by FY30 and c.285 GWh by FY35 on our base-case estimates, but domestic cell output is barely 1–2 GWh.
 - **Where India is localizing: vehicle assembly, battery packs/BMS, e-motors and e-drives (40–60% local in 2W/3W), and increasingly cell assembly. Where it still depends heavily: cells, cathode/anode/electrolyte/LiPF₆ (near-zero commercial output), rare-earth magnets, power semiconductors and ADAS/cockpit silicon.** The policy architecture (ACC PLI, PM E-DRIVE, NCMM, rare-earth magnet scheme, SPMEPCI, semiconductor mission) is largely in place, but midstream processing and cell manufacturing will take c.5–7 years to mature — leaving India exposed through the steepest part of its EV ramp.
 - **OEMs are managing risk through vertical integration (Ola, Tata/Agratas, Amara Raja), Chinese tech licensing (Exide–SVOLT, Amara Raja–Gotion, Mahindra–BYD), multi-sourcing and rare-earth-free motor development (Sona Comstar, TVS, Chara). The 2025 Chinese rare-earth episode — which cut Bajaj Chetak output 47% YoY in July 2025 — was the wake-up call.** India's path rhymes with China's a decade ago but will be slower, less vertically integrated, and shaped by Press Note 3 FDI curbs that block the very Chinese JVs that could accelerate it.
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Page 1: Executive Summary

India's EV market has reached genuine scale, but the supply chain underneath it is largely borrowed. In FY26 India sold 2,452,921 EVs (Vahan retail), up 25% YoY from 1,967,547 in FY25, taking penetration to c.8.3% of all vehicle registrations. Electric two-wheelers (1.40m units, 57% of EV volumes) and three-wheelers (831k units, 34%) dominate volumes; passenger EVs nearly doubled to c.200k units (8% share) and commercial EVs remain nascent. Yet almost every cell powering this fleet is imported — c.75% of Li-ion cells in Indian BEVs come from China, and India is c.100% import-dependent for lithium, cobalt and nickel and c.93% for the rare-earth magnets in EV motors. India assembles packs and builds vehicles; China makes the chemistry.

Traction battery demand is set to grow roughly 5x by FY30 and c.14x by FY35. On our base-case segment build-up, India's automotive traction battery demand rises from c.20 GWh in FY26 to c.95–100 GWh by FY30 and c.285 GWh by FY35, driven first by two- and three-wheelers (small packs, high penetration) and later by passenger vehicles and buses (large packs, slower penetration but far higher kWh per unit). The critical investment implication is timing: the demand curve steepens from FY27 onward, precisely when domestic cell capacity is still ramping. As of October 2025, only 2.8% (1.4 GWh) of the ACC PLI's 50 GWh target had been commissioned — entirely by Ola Electric — against an ambition of 50 GWh by 2025.

Where India is localizing, and where it is not, is now fairly clear. Vehicle assembly, battery-pack integration and BMS, and e-motors/e-drives are the bright spots — motor and controller localization runs 40–60% in 2W/3W, and firms like Sona Comstar, Bosch India, Tata AutoComp and Bharat Forge [Knowledge-sourcing](#) have real capability. Cell manufacturing is finally moving from pack assembly to true cells at Ola (Krishnagiri), and Agratas (Tata), Amara Raja and Reliance are building tens of GWh for commercial output in 2026–2027. But the midstream — cathode active material, anode/graphite, separators, electrolyte and LiPF₆ — is almost entirely absent at commercial scale, and rare-earth magnets, power semiconductors (IGBT/MOSFET/SiC) and ADAS/cockpit silicon remain near-total import dependencies. India's cells will cost a 15–20% premium to Chinese imports for years, because the inputs are still Chinese.

OEMs are managing supply risk in four ways, and the 2025 rare-earth shock forced the issue. First, vertical integration — Ola's in-house 4680/46100 Bharat Cell, Tata's Agratas, Amara Raja's Gigacorridor. Second, Chinese technology licensing — Exide–SVOLT, Amara Raja–Gotion, Mahindra's BYD Blade cells — which is fast but hostage to Beijing's export licensing. Third, multi-sourcing and pack-level control — Tata sources cells from Gotion and EVE via Tata AutoComp and Octillion; Hyundai and Maruti localize pack assembly while importing cells. Fourth, engineering around the bottleneck — Sona Comstar, TVS and start-up Chara Technologies have developed rare-earth-free motors after China's April 2025 controls on seven rare-earth elements (which caused a 75% drop in global magnet exports) cut Bajaj's Chetak output 47% YoY in July 2025. When Beijing's Decision No. 58 threatened in late 2025 to place complete cells, cathodes and anode-graphite under export licensing, it demonstrated a veto over India's entire battery ramp.

Our bottom line for investors: India's EV volume story is investable; its supply-chain-independence story is a 2030s story, not a 2020s one. The nodes that matter most given the GWh outlook — in order — are cell manufacturing, midstream cathode/anode/electrolyte processing, rare-earth magnets, and power electronics. Progress here is real but slow, and heavily dependent on Chinese equipment, technicians and precursors that Press Note 3 makes hard to bring onshore via JV. India will look like a slower, less vertically integrated China, hedged by friend-shoring (Argentina, Australia, Chile, the Quad Minerals partnership, US-India TRUST/ICET) rather than domestic self-sufficiency. Component suppliers with genuine IP (Sona Comstar) and vertically integrating OEMs (Tata, Ola) are the cleaner ways to play localization; pure pack-assemblers carry the most import/FX/geopolitical risk.

Key Findings

- **Demand is 2W/3W-led today but PV/bus-led in GWh by FY28.** FY26 EV mix: e-2W 57%, e-3W 34%, e-PV 8%, e-CV <1%. But because a PV pack (30–79 kWh) is c.10–25x a 2W pack (2–5 kWh) and a bus pack (250–380 kWh) is c.100x, PVs and buses drive most incremental GWh from FY28.
- **Cell manufacturing has failed on timeline but is finally moving.** ACC PLI delivered 2.8% of target; the real action is non-PLI (Agratas 20 GWh, Amara Raja 16 GWh, Reliance 30–40 GWh, Exide 6 GWh) commissioning in 2026–27, plus Ola's operational c.2.5 GWh.
- **The midstream is the deepest hole.** India has essentially no commercial-scale cathode (CAM), synthetic-graphite anode, Li-ion separator or electrolyte production. The lone bright spot is GFCL EV's LiPF₆ electrolyte-salt plant (c.1,800 tpa Phase 1), the first commercial-scale battery chemical in the country.
- **Magnets are the sharpest single-point-of-failure.** India imported 53,748 tonnes of NdFeB magnets in FY25 (c.\$291m per SBI), c.93% from China. The ₹7,280 cr scheme targets 6,000 MTPA but output is a 2028+ event.
- **Motors and e-drives are India's strongest node**, with Sona Comstar producing 180,000+ indigenous traction motors and leading rare-earth-free (ferrite-assisted SynRM) development.

Details

2. India's EV demand and battery needs

India's EV volumes are dominated by 2W and 3W today, but GWh will increasingly be driven by PVs and buses. All FY24–FY26 figures are Vahan retail registrations (which differ from SIAM wholesale dispatches — SIAM does not publish granular EV splits, and its PV/2W/3W dispatch totals run higher than Vahan retail; e.g. SIAM FY25 2W dispatches were 19.6m vs Vahan-based retail lower). FY27–FY35 are Bernstein estimates. Pack sizes are Bernstein segment averages built from model-level disclosures.

Table 2.1 — Electric two-wheelers (e-2W)

FY	Total 2W sales (m)	EV pen. (%)	EV volume (m)	Avg pack (kWh)	Battery demand (GWh)
FY24 (A)	17.7	5.4%	0.95	2.8	2.7
FY25 (A)	19.6	6.2%	1.15	2.9	3.3
FY26 (A)	21.4	6.6%	1.40	3.0	4.2
FY27 (E)	22.5	9.5%	2.1	3.1	6.5
FY30 (E)	26.0	22%	5.7	3.2	18.3
FY35 (E)	32.0	50%	16.0	3.4	54.4

Assumptions: mass e-2W packs 2.0–3.5 kWh, premium 4–5.2 kWh (e.g. Ola S1 Pro+ 5.2 kWh). Penetration ramps as TCO parity holds post the September 2025 GST cut. FY26 monthly run-rate: 7.8% (Apr'26) rising to 11.2% (Jul'26) per Vahan.

Table 2.2 — Electric three-wheelers: passenger (e-rickshaw + L5 passenger)

FY	Total 3W pax (m)	EV pen. (%)	EV volume (m)	Avg pack (kWh)	Battery demand (GWh)
FY24 (A)	~1.0	~48%	0.48	5.5	2.6
FY25 (A)	~1.05	~54%	0.57	5.8	3.3
FY26 (A)	~1.05	~57%	0.60	6.0	3.6
FY27 (E)	1.10	62%	0.68	6.2	4.2
FY30 (E)	1.25	78%	0.98	6.5	6.4
FY35 (E)	1.45	92%	1.33	7.0	9.3

Note: e-rickshaws (low-speed, 3–5 kWh, transitioning from lead-acid to Li-ion) plus L5 passenger autos (7–8 kWh). FY26: 477,765 e-rickshaws; L5 passenger up 76% YoY. L5 passenger EV penetration hit 52.6% in Jul'26 (Vahan). Battery-swapping standardised packs prevalent here.

Table 2.3 — Electric three-wheelers: cargo (e-cart + L5 cargo)

FY	Total 3W cargo (k)	EV pen. (%)	EV volume (k)	Avg pack (kWh)	Battery demand (GWh)
FY24 (A)	~350	~22%	77	7.0	0.5
FY25 (A)	~380	~28%	106	7.5	0.8
FY26 (A)	~400	~33%	130	8.0	1.0
FY27 (E)	430	40%	172	8.2	1.4

FY30 (E)	520	62%	322	8.5	2.7
FY35 (E)	650	85%	553	9.0	5.0

FY26: 86,466 e-carts. e-Goods carrier (N-category) up 166.9% YoY to 14,803 units.

Table 2.4 — Electric passenger vehicles (e-PV / e-4W)

FY	Total PV sales (m)	EV pen. (%)	EV volume (k)	Avg pack (kWh)	Battery demand (GWh)
FY24 (A)	4.22	1.9%	80	34	2.7
FY25 (A)	4.30	2.4%	101	38	3.8
FY26 (A)	4.35	4.6%	200	42	8.4
FY27 (E)	4.55	7%	319	45	14.4
FY30 (E)	5.20	18%	936	50	46.8
FY35 (E)	6.20	40%	2,480	55	136.4

e-PV registrations reached c.193,600–222,870 in FY26 (source basis varies: Vahan e-4W 193,633; EVreporter 222,870 incl. quadricycles), up c.90% YoY. Pack assumptions: entry (Tiago.ev ~19–24 kWh, Punch.ev ~25–35 kWh), mid (Nexon.ev 30/45 kWh, Creta Electric 42/51.4 kWh NMC, e Vitara 49/61 kWh LFP, MG Windsor/ZS), premium (Harrier.ev 65/75 kWh LFP, Mahindra BE 6/XEV 9e 59/79 kWh Blade LFP). Tata c.39% e-PV share FY26; Mahindra +443% YoY; JSW MG +83%.

Table 2.5 — Electric LCV (N1 goods)

FY	Total LCV (k)	EV pen. (%)	EV volume (k)	Avg pack (kWh)	Battery demand (GWh)
FY24 (A)	~600	~0.6%	4	25	0.1
FY25 (A)	~610	~1.0%	6	27	0.2
FY26 (A)	~620	~1.6%	15	28	0.4
FY27 (E)	640	3%	19	30	0.6
FY30 (E)	720	12%	86	35	3.0
FY35 (E)	850	35%	298	40	11.9

Table 2.6 — Electric M&HCV / trucks

FY	Total M&HCV (k)	EV pen. (%)	EV volume (k)	Avg pack (kWh)	Battery demand (GWh)
FY24 (A)	~350	~0.1%	0.4	200	0.1
FY25 (A)	~355	~0.2%	0.7	220	0.2
FY26 (A)	~360	~0.5%	1.8	240	0.4
FY27 (E)	375	1.5%	5.6	250	1.4
FY30 (E)	430	6%	26	300	7.7

FY35 (E)	520	22%	114	350	40.1
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e-truck adoption anchored by PM E-DRIVE's ₹500 cr truck allocation; still pilot-stage.

Table 2.7 — Electric buses (9m/12m)

FY	Total buses (k)	EV pen. (%)	EV volume (k)	Avg pack (kWh)	Battery demand (GWh)
FY24 (A)	~55	~6%	3.4	250	0.9
FY25 (A)	~60	~8%	4.8	300	1.4
FY26 (A)	~65	~11%	7.2	320	2.3
FY27 (E)	70	18%	12.6	340	4.3
FY30 (E)	85	40%	34	360	12.2
FY35 (E)	100	70%	70	380	26.6

PM E-DRIVE funds 14,028 e-buses (₹4,391 cr); 13,800 sanctioned across Bengaluru (~4,500), Delhi (2,800), Hyderabad (2,000), Ahmedabad (1,000), Surat (600) [Anantam IAS](#) etc. Olectra-BYD, Tata, JBM, Switch, VECV key suppliers.

Table 2.8 — Aggregate traction battery demand (GWh), base case

FY	2W	3W pax	3W cargo	PV	LCV	M&HCV	Bus	Total (GWh)
FY24 (A)	2.7	2.6	0.5	2.7	0.1	0.1	0.9	~9.6
FY25 (A)	3.3	3.3	0.8	3.8	0.2	0.2	1.4	~13.0
FY26 (A)	4.2	3.6	1.0	8.4	0.4	0.4	2.3	~20.3
FY27 (E)	6.5	4.2	1.4	14.4	0.6	1.4	4.3	~32.8
FY30 (E)	18.3	6.4	2.7	46.8	3.0	7.7	12.2	~97
FY35 (E)	54.4	9.3	5.0	136.4	11.9	40.1	26.6	~284

Scenario framing. Our base case reaches c.97 GWh of automotive traction demand by FY30. Adding stationary storage and consumer electronics, NITI Aayog's all-sector figure is c.600 GWh cumulative 2022–2030; IEEFA cites c.272 GWh annual demand by FY30; WRI India c.160 GWh by 2030; NITI Aayog/GGEF put cumulative EV-battery demand at c.381 GWh by 2030. Our bull case (faster PV/bus penetration) reaches c.130–150 GWh automotive by FY30; our bear case (subsidy cliff, slower PV) c.75 GWh. The through-line: PVs and buses, though modest in unit terms, drive most incremental GWh from FY28, exactly when the cell build-out must land.

3. Mapping India's EV supply chain — batteries and upstream

India controls the downstream (pack, BMS, vehicle) and almost none of the upstream chemistry.

Table 3.1 — Battery value-chain localization

Node	Local content (est.)	Key domestic players	Main import source	Resilience
Lithium (raw/refined)	~0%	KABIL (overseas), GSI exploration	Chile, China, Ireland (carbonate)	Critical risk

Cobalt	~0%	Recyclers (Lohum, Attero)	China (processed), DRC	Critical risk
Nickel	~0%	—	Australia (65% of oxides), China	Critical risk
Natural graphite	~40% mined, ~0% battery-grade	Himadri (feedstock)	Mozambique, China	High risk
Synthetic graphite (anode)	~0% commercial	Epsilon (2,500tpa+pilot), Himadri (200 MTPA)	China (c.91% FY25)	Critical risk
Cathode (LFP/NMC CAM)	~0% commercial	Altmin (FS stage), Himadri (40 ktpa announced), Epsilon (10 ktpa 2026)	China	Critical risk
Separator	~0% (Li-ion)	Daramic (lead-acid only)	China, Japan, Korea	Critical risk
Electrolyte	Early	Neogen, GFCL EV	China	High risk
LiPF6 (salt)	Low (first plant live)	GFCL EV (~1,800 tpa Phase 1)	China	High risk
Cell	~5% (c.1–2 GWh of ~25 GWh)	Ola, Agratas, Amara Raja, Exide, Reliance	China (c.75%)	Critical risk
Pack + BMS	60–80%	Tata AutoComp, Amara Raja, TDSG, OEM in-house	(cells imported)	Moderate
Cell-making equipment	~0%	—	China, Korea, Japan	High risk

Table 3.2 — India cell capacity: ACC PLI vs non-PLI

Company	Location	Announced GWh	Chemistry	Status	Commissioning
ACC PLI					
Ola Electric	Krishnagiri, TN	20 (5 near-term)	NMC/LFP (4680/46100)	c.2.5 GWh operational; c.6 GWh installed	FY25 onward
Reliance New Energy	Jamnagar, GJ	10 (+5 R1)	LFP	Land secured; on-time claim	2026–27
Rajesh Exports	Karnataka	5	—	Minimal progress	Delayed
(Hyundai Global)	—	20	—	Withdrew; re-tendered 10, reserved 10	Cancelled
Non-PLI					
Agratas (Tata)	Sanand, GJ	20 (Ph1; 40 total)	NMC (AESC lic.); LFP pilot in-house	Steel frame complete Apr'26	2027
Amara Raja	Mahbubnagar, TS	16	NMC then LFP (Gotion lic.)	1.5 GWh pack live; cell deferred	Q2–Q3 FY27

Exide Energy	Bengaluru, KA	6 (Ph1)	LFP/NMC (SVOLT lic.)	Trial production	FY26 end
Reliance	Jamnagar	30–40 (to 100)	LFP	Construction	2026–27
TDSG (Suzuki-Toshiba-Denso)	Hansalpur, GJ	~30m cells/yr	LTO/NMC (hybrids)	Operational (hybrids); EV line pending	EV FY27

ACC PLI: ₹18,100 cr outlay; 40 GWh awarded, no incentives disbursed; show-cause notices issued; most beneficiaries seeking 18-month extensions. Only Exide and Amara Raja had prior battery experience — and neither qualified in either auction round. Ola scaled ambition back to 5 GWh by FY29.

Table 3.3 — Cell cost: imported Chinese vs Indian-made (BloombergNEF 2025 survey, 9 Dec 2025)

Metric	Value
Global avg pack 2025	\$108/kWh (–8% YoY, record low)
China avg pack 2025	\$84/kWh (–13% YoY)
BEV pack / LFP pack / NMC pack	\$99 / \$81 / \$128 per kWh
China LFP cell (low) 2025	\$44–60/kWh; lowest observed cell/pack \$36/\$50
2W/3W pack price 2025	\$133/kWh
Landed Chinese cell into India	c.\$85/kWh (IEEFA)
Indian-made cell premium (est.)	+15–20% (Bernstein)
2026 trend	Modest further decline expected; lithium-carbonate up >90% off June’25 trough poses upside risk

Table 3.4 — Mineral import dependence

Mineral	India import dependence	Top source	Import data point
Lithium (all)	~100%	Chile, China, Ireland	Li carbonate 1,000t/\$12m FY25 (Ireland 40%); Li oxide/hydroxide 2,000t/\$23m (Chile 41%, China 38%)
Cobalt	~100%	China (processed), DRC	—
Nickel	~100%	Australia (65% of oxides/hydroxides)	—
Natural graphite	~60%	Mozambique (overtook China)	prices \$2,000→\$1,271/t (Jun’23→Oct’25)
Synthetic graphite	~100%	China (c.91% FY25)	57,000t / ~\$58m
Li-ion	~75% (BEVs)	China	>\$3bn FY25 (8x FY19’s \$384m)

cells			
Critical mineral bill	—	—	\$3.03bn (FY21) → \$8.01bn (FY24)

Domestic exploration: Katghora (Chhattisgarh) lithium/REE block auctioned successfully; Reasi/Salal-Haimna (J&K) annulled twice for no/low bids (clay-hosted, untested extraction). Seventh tranche: nine blocks scrapped for poor response; sixth tranche: eleven blocks cancelled. Auctions to date: c.34+ blocks awarded across tranches, few reaching production. NCMM ₹34,300 cr. KABIL secured 5 lithium blocks in Catamarca, Argentina.

Table 3.5 — Recycling capacity and offset scenario

Metric	2026	2030 (E)	2035 (E)
Operational Li-ion recycling capacity	~2 GWh throughput	~30–50 GWh	~100+ GWh
Registered recyclers (CPCB EPR portal)	487 (4,022 producers)	—	—
End-of-life available for recycling	small	~128 GWh (NITI)	7x 2030
Realistic mineral demand offset	<5%	~15–20%	~28–43% (NITI/GGEF)

Key players: Lohum (c.70% share, 20,000→50,000 tpa, 1,000 MTPA Li refinery), Attero (c.15,000 tpa Li-ion → 200,000 tpa planned + AP plant), Tata Chemicals, RecycleKaro (95%+ recovery, 99.5% purity CoSO₄/NiSO₄), BatX, Exigo. BWMR 2022/EPR framework live; only 30–40% of end-of-life batteries flow through formal channels; black-mass export curbs tightened; Critical Mineral Recycling Incentive Scheme ₹1,500 cr with 58 recyclers allotted. Lead-acid recovery target 60% by FY27.

4. Motors, magnets, power electronics, BMS and the wider EV tech stack

Motors/e-drives are India’s strongest localization node; magnets and semiconductors are its weakest. PMSMs account for 70–80% of EV traction applications and are the rare-earth choke-point.

Table 4.1 — Master localization: EV tech stack

Node	Current local %	Main Indian/JV players	Foreign dependency (source)	FY30 (E)	FY35 (E)
E-motor (2W/3W)	40–60%	Sona Comstar (180k+ motors), Bosch India, Tata AutoComp, Electrodrive	Magnets (China)	65%	80%
E-motor (PV)	20–40%	Sona Comstar, Valeo India, Bharat Forge	Valeo/BorgWarner design; magnets	45%	65%
E-axle / e-drive	30–50%	Sona Comstar, Tata AutoComp	ZF, Nidec, BorgWarner	55%	70%
Inverter / controller	30–50%	Sona Comstar, Bosch, Napino	Power semis inside	50%	65%
NdFeB magnets	~0%	Midwest Advanced Materials (pilot), IREL (feedstock)	China (c.93%)	10–15%	30–40%
IGBT/MOSFET/SiC	~0% fab	(CG-Renesas OSAT packaging, auto-grade)	Infineon, ST, onsemi (DE/US/CN)	~0% fab	5–10%

Onboard charger	40–60%	Delta India, Exicom, Servotech, Napino, Minda	Semis inside	65%	80%
DC-DC converter	40–60%	Minda, Sona, Varroc	Semis inside	65%	80%
BMS hardware	40–60%	Napino, Minda, Ionage, TVS/Ather in-house	ICs (TI, NXP, ADI)	60%	75%
BMS firmware	60–80%	Ather, Ola, TVS, KPIT, Tata Elxsi	—	80%	90%
Thermal management	40–60%	Subros, Tata AutoComp	Compressor ICs	60%	75%
Domain controller / telematics (AIS-140)	20–40%	KPIT, Tata Elxsi, Bosch	SoCs (Qualcomm, NXP)	40%	60%
ADAS SoC	~0%	(design: KPIT, LTTS)	Nvidia, Qualcomm, Mobileye	~0%	5%
ADAS software	30–50%	KPIT, Tata Elxsi, Bosch India, LTTS	—	60%	75%

L2 ADAS is penetrating Indian PVs (Mahindra XEV 9e uses Mobileye + Qualcomm; most premium EVs use Qualcomm Snapdragon cockpit SoCs) — mirroring China’s dependence on Qualcomm cockpit / Nvidia ADAS silicon, but India adds no domestic SoC alternative. Rare-earth-free motor programmes: Sona Comstar (ferrite-assisted SynRM, validated Oct 2025), TVS (locally-sourced larger magnets), Chara Technologies (SynRM, 20% cost edge), Vimag Labs (VMSM patent, unproven at scale).

Table 4.2 — Rare-earth magnet supply

Metric	Value
India NdFeB imports FY25	53,748 tonnes (c.\$291m per SBI)
China share of India’s imports	~93% (Volza)
China share of global NdFeB output	~85% (85kt of 100kt)
REPM scheme outlay	₹7,280 cr (\$815m), Cabinet-approved Nov 2025
Split	₹6,450 cr sales-linked + ₹750 cr capital subsidy
Target	6,000 MTPA across ≤5 firms (≤1,200 MTPA each), 7-yr (incl. 2-yr gestation)
Bidding	RFP 20 Mar 2026; pre-bid 7 Apr; bids due 28 May 2026
Interested	Vedanta, JSW, Sona Comstar, Bharat Forge, Uno Minda, Lynas, Iluka, Rainbow, Midwest, Proterial
IREL role	Assured NdPr-oxide supply to 3 lowest bidders
Realistic meaningful output	2028–2030

Table 4.3 — India semiconductor fab/OSAT timeline (EV-relevant)

Facility	Type	Status	Timeline
Micron, Sanand	ATMP (memory)	Operational (inaugurated 28 Feb 2026)	Live
Kaynes Semicon, Sanand	OSAT	Commercial (Mar 2026)	Live
CG Power–Renesas–Stars, Sanand	OSAT (auto-grade AEC-Q100)	Commercial (4 Jul 2026)	Live
Tata Electronics, Jagiroad (Assam)	ATMP	Operational	Live
Tata–PSMC, Dholera	Fab (28nm, 50k wpm)	Construction >50%; ASML tools secured	First silicon H1 2027 (slipped)
Budget 2026-27 semiconductor outlay	—	₹8,000 cr (largest single-year)	—

OSAT = packaging/testing only; wafers still imported. >90% of fab equipment and 85–90% of specialty chemicals imported. Dholera’s 28nm node is the sweet-spot for automotive/power-electronics chips but is a 2027+ (realistically 2030s at scale) EV-relevance story.

5. How Indian OEMs are managing the supply chain today

Table 5.1 — OEM supply-chain matrix

OEM	Cell sourcing	Motor/e-drive	Power electronics & BMS	Magnet exposure	Vertical integration	Key partnerships	Response to China curbs
Tata Motors/TPEM	Gotion & EVE (China) via Tata AutoComp & Octillion (Pune, 2 GWh); → Agratas 2027	Tata AutoComp, in-house	Tata AutoComp, in-house BMS, Tata Elxsi	High (PMSM)	High (Agratas, AutoComp, Elxsi)	Agratas–AESC (NMC); LFP in-house	Building Agratas; forced LFP pivot to own tech after China licensing blocked
Mahindra	BYD Blade LFP (FinDreams, imported); VW unified cells (2026 planned)	Valeo (3-in-1)	Bosch VCU, Sibros OTA, Qualcomm MAIA	High	Medium	BYD, VW, Valeo, BorgWarner, Mobileye, Qualcomm	Multi-source; VW cell deal as hedge
Maruti Suzuki	BYD LFP (e Vitara 49/61 kWh); TDSG (hybrids)	Suzuki/in-house	In-house	Medium	Medium (TDSG JV)	Toshiba–Denso–Suzuki (TDSG)	Export-led (2,900+ units to Europe); localizing packs FY27
	NMC imported		Hyundai-				Local pack assembly;

Hyundai India	(Creta Electric 42/51.4 kWh); Exide LFP offtake (planned)	supplier	Mobis JV pack (75k/yr, Chennai)	Medium	Medium	Exide, Mobis	92% overall localization; Exide LFP deal
JSW MG	Chinese (MG/SAIC); Windsor/ZS	supplier	supplier	High	Low	JSW-SAIC JV	JSW bidding for rare-earth & possibly cell schemes
Bajaj Auto	Imported cells	In-house motors	In-house	Very high — Chetak output –47% YoY Jul’25	Medium	—	Switched to light-REE magnets; developing REE-free; delayed entry Chetak
TVS Motor	Imported cells	In-house	In-house	High	Medium	—	Locally-sourced larger magnets; REE-free R&D; least disrupted
Hero/Ather	Imported cells; in-house packs	In-house	In-house BMS/firmware	High	Medium (Ather packs)	—	Ather faced order-fulfilment gaps; Hero secured Q2 supply
Ola Electric	In-house 4680/46100 Bharat Cell (Krishnagiri, ~2.5 GWh live)	In-house	In-house	High	Highest (cell-to-vehicle)	—	Vertical integration is the hedge; also home BESS (Shakti)
Ashok Leyland/Switch	Imported/pack	Switch in-house	supplier	High	Medium	—	—
JBM	Imported/pack	supplier	supplier	High	Low	—	e-bus focus
Olectra	BYD (China)	BYD	BYD	High	Low	BYD JV	Most exposed to China

Table 5.2 — Risk management by segment

		Magnet		
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Segment	Cell strategy	exposure	Integration	Key risk
2W	In-house packs; Ola in-house cells, others import	High (small PMSM)	Rising (Ola, Ather)	Magnet + cell import
3W	Imported cells, LFP packs; swapping standard	Medium	Low-medium	Subsidy cliff / price
PV	BYD/Chinese cells → domestic 2027; local packs	High	Medium (Tata)	Cell + magnet + FX
CV/bus	Large imported LFP; some BYD	High	Low	Financing + China cell

6. Policy push vs execution reality

Table 6.1 — Targets-vs-actuals scorecard

Scheme	Announced target	Outlay	Actual to date	Slippage	Bernstein assessment
ACC PLI	50 GWh cells by 2025	₹18,100 cr	1.4 GWh commissioned (Ola only) as of Oct'25; 40 GWh awarded; nil incentive disbursed	Severe (2.8%)	Failed on timeline; extensions/relaxations under discussion
NCMM	Exploration, processing, overseas	₹34,300 cr	Auctions run; many blocks unsold; REE corridors announced	Moderate	Right architecture, weak execution
PLI-Auto	Champion OEM + component	₹25,938 cr	Disbursements ramping; DVA/PMP thresholds	Moderate	Slow start, improving
PM E-DRIVE	24.79 lakh e-2W, 14,028 e-buses	₹10,900 cr	22.12 lakh EVs (19.19 lakh e-2W + 2.93 lakh e-3W); ₹1,703 cr reimbursed	FY27 budget cut to ₹1,500 cr	Front-loaded; demand now largely TCO-driven
FAME-II	1.6m EVs	~\$1.38bn	Lapsed Mar 2024; broadly met 2W/3W	—	Succeeded on 2W/3W
SPMEPCI	Global OEMs (15% duty on \$35k+ CBUs, ₹4,150cr inv, 25%→50% DVA)	duty forgone (cap \$759m/applicant)	Zero applications at first lapse (Oct'25); reopened/revised	Severe	Tesla declined (import-only); VW/Merc/Skoda/Hyundai/Kia interested; BYD barred (Press Note 3)
Rare-earth magnet scheme	6,000 MTPA	₹7,280 cr	RFP Mar'26, bids due May'26	Early	Promising; 2028+ output
	Exempt		Li/cobalt/critical		

Customs/Budget	critical minerals & capital goods	—	minerals exempted (Budget 2024-25); REE corridors (Budget 2026-27)	—	Helpful at margin
KABIL	Overseas mineral assets	—	5 Li blocks Argentina (Catamarca); MoUs Australia/Chile/Zambia	On track	Slow to first output
Battery waste/EPR	Collection & recycling targets	—	BWMR 2022 live; 487 recyclers; CMRIS ₹1,500 cr	On track	Framework ahead of capacity

Table 6.2 — Bottleneck severity

Bottleneck	Severity	Time-to-resolve
Midstream processing (CAM/anode/electrolyte)	Critical	5–7 yrs
Chinese equipment & technician access (visas)	High	3–5 yrs (Press Note 3)
Rare-earth magnet capacity	High	3–4 yrs
Skills/talent	High	3–5 yrs
Critical mineral financing / reserve data	Medium-high	ongoing
Testing infrastructure	Medium	2–3 yrs
Permitting (mining)	Medium	ongoing

7. Geopolitics, resilience, and investment implications

Table 7.1 — Risk-exposure matrix

Node	India exposure	China's control	Available hedge
Lithium raw	~100% import	Refining c.60%	KABIL (Argentina), Australia, Chile
Graphite/anode	~100% synthetic	c.91–97% anodes	Mozambique graphite; Epsilon/Himadri
Cathode (CAM)	~100%	dominant	Altmin, Himadri, Epsilon (2026–31)
Cells	c.75%	c.77% global	Ola, Agratas, Amara Raja, Reliance
Cell equipment	~100%	dominant	Korea/Japan; Press Note 3 blocks China
NdFeB magnets	c.93%	c.85–93%	REPM scheme, REE-free motors, IREL
Power semis	~100%	growing	Infineon/ST/onsemi; ISM (2027+ fab)
ADAS/cockpit SoC	~100%	(US-led)	Qualcomm/Nvidia (US); design in India only

The key risks for investors are concentrated, sequenced and largely outside India's control. The single biggest is China's ability to throttle the battery and magnet upstream at will — demonstrated by the April 2025 rare-earth controls (which caused a 75% drop in global magnet exports and cut Bajaj Chetak output 47% YoY in July 2025), the July 2023 graphite controls, and the late-2025 Decision No. 58 threat over complete cells, cathodes and anode-graphite (suspended for one year to November 2026 after US–China talks). Because Press Note 3 restricts Chinese FDI, India

cannot easily import the JV model that built China's chain — Chinese firms can license technology (Gotion, SVOLT, BYD) but cannot take equity, and even licensing now runs into Beijing's export-control net, as Agratas found when forced to develop LFP in-house.

The unlock areas that matter most, ranked by how much they move the GWh needle: (1) cell manufacturing, (2) midstream cathode/anode/electrolyte, (3) rare-earth magnets, (4) power electronics. Cell manufacturing matters most because it is the largest value pool and the biggest FX/geopolitical exposure — every GWh of the c.97 GWh FY30 demand not made domestically is imported dependency, and at c.\$85/kWh that is a multi-billion-dollar annual import bill (IEEFA estimates >\$23bn/yr by FY30 if demand is met entirely by imports). But cells without midstream are only half-localized: a domestic cell using Chinese cathode and anode is still Chinese at its core, which is why the CAM/anode/electrolyte gap (near-zero commercial output today) is the deeper structural problem. Magnets are a smaller value pool but a sharper single-point-of-failure, as Bajaj showed. Power electronics is the longest-dated: OSAT packaging is live, but automotive-grade fab output is a 2027+ and, realistically, 2030s story.

India's trajectory rhymes with China's a decade ago, but it will be slower, less vertically integrated, and more externally hedged. China localized by combining massive domestic demand, state-directed capital, vertical integration and unrestricted access to its own upstream — reaching >90% localization across cells, cathodes, anodes, separators and e-drives, holding c.77% of global cell output and 80–100% of global capacity in cathodes/anodes/separators/electrolytes. India has the demand and increasingly the policy, but lacks the midstream, the mineral base, the equipment industry and — critically — the ability to partner freely with the one country that has all three. India's realistic path is a hybrid: domestic cell assembly and pack/motor localization on top of a friend-shored mineral and midstream layer (Argentina/Australia/Chile lithium, Quad Minerals Security Partnership, US–India TRUST/iCET), with genuine self-sufficiency in magnets and midstream chemicals arriving only in the 2030s.

Recommendations

Stage 1 (now–FY27): play volume and localization winners; avoid pure importers. The cleanest exposures are component suppliers with real IP and export franchises (Sona Comstar in e-drives and rare-earth-free motors), vertically integrating OEMs (Tata Motors via Agratas; Ola Electric if execution and quality hold), and pack/BMS/power-electronics names with rising DVA (Amara Raja, Exide, Uno Minda, Napino). The highest-risk profiles are OEMs most exposed to Chinese cells and magnets without a hedge (Olectra-BYD; pure 2W/3W players on imported PMSM motors). *Benchmark that would change this: a second Chinese magnet/graphite export freeze before November 2026 would sharply favour the rare-earth-free-motor and vertically integrated names and punish importers.*

Stage 2 (FY27–FY29): watch the cell commissioning cadence as the key catalyst. Track whether Agratas (target 2027), Amara Raja (Q2–Q3 FY27) and Reliance actually commission on time and hit DVA thresholds. Successful ramp de-risks the FX/import thesis and re-rates the vertically integrated OEMs; further slippage (the base-rate outcome given ACC PLI's 2.8% record) keeps India import-dependent and keeps the cost premium. *Benchmark: >20 GWh of genuinely commissioned domestic cell capacity by end-FY28 would be the signal that localization is real; <10 GWh confirms the bear case.*

Stage 3 (FY29–FY35): position for the midstream and magnet unlock — but not before the evidence. Epsilon, Altmin, Himadri (CAM/anode), GFCL EV (LiPF6) and the REPM-scheme winners (Vedanta, JSW, Sona Comstar) are the long-dated optionality. Do not underwrite meaningful revenue before FY29 for midstream or FY28 for magnets. *Benchmark: first commercial-scale (>10 ktpa) Indian cathode output, and first >1,000 MTPA domestic NdFeB magnet output, are the two datapoints that would justify moving these from optionality to core holdings.*

Cross-cutting: monitor policy execution, not announcements. The gap between announced and delivered is India's defining supply-chain feature. Weight MHI Parliament answers, CPCB EPR portal data and DGCIS trade data far above press releases.

Caveats

- **Actuals vs estimates.** FY24–FY26 vehicle and EV figures are Vahan retail registrations; FY27–FY35 are Bernstein estimates. Pack sizes and segment GWh from FY27 are Bernstein assumptions, labelled in every table. Our FY30 automotive base case (c.97 GWh) is deliberately below headline all-sector forecasts (NITI 600 GWh cumulative; IEEFA 272 GWh; NITI/GGEF 381 GWh EV-battery) because those include stationary storage, consumer electronics and/or cumulative figures.
- **Vahan vs SIAM.** Vahan captures registrations (retail); SIAM captures wholesale dispatches and does not publish granular EV splits. Totals differ — e.g. SIAM PV FY25 4.3m dispatches vs lower Vahan retail. We use Vahan for EV granularity and flag where SIAM differs. Readers should cross-check via the Vahan analytics dashboard (parivahan), SIAM flash reports and FADA retail data.
- **Data not directly obtainable.** For precise import volumes/values by node, consult DGCIS/Ministry of Commerce trade data by HS code (HS 850760 Li-ion cells/packs, 850790 battery parts, 280530 rare-earth metals, 850511 permanent magnets, 250410 graphite). For scheme status, MHI Parliament answers and PIB releases. For OEM DVA and cell sourcing, company annual reports. EVreporter's paid FY26 report holds the fullest segment-level registration dataset.
- **Sourcing quality flags.** The Agratas in-house LFP pivot (4 Aug 2026) rests on a Bloomberg report citing unnamed sources; the NMC-by-2027 and 20 GWh Phase 1 details are well-corroborated. Tata's cell-supplier granularity (Gotion/EVE via Tata AutoComp/Octillion) is from trade/enthusiast outlets and is directionally reliable (all point to Chinese cells today) but exact cell models are indicative. Maruti e Vitara's confirmed launch cell source is BYD LFP; multi-sourcing (TDSG/others) is prospective. NdFeB import tonnage (53,748t FY25) and c.93% China share come from government data and Volza trade intelligence respectively; treat as best-available rather than audited.
- **Forward-looking language.** Where sources use "could", "targeted", "expected" or give projections (e.g. IESA's 10m EV sales by 2030, CareEdge's cell-import-dependence falling to 20% by FY27), we have flagged them as forecasts, not facts. Scheme outlays are authorised amounts, not disbursements — actual ACC PLI incentive disbursement to date is nil.